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The Solidarity of Premier league teams

Do soccer teams that have played together for a longer period achieve greater league success than younger teams?

Analysis by

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Abstract

If you follow the Premier League or any top soccer league for that matter you can't help but notice the amount of player transfers that happen between teams. Players seem to move quite frequently and short stays are more common than long ones. This is a correct observation, just by looking at the 2016-'17 Premier league season there were 224 individuals playing for a new team. This is in a year when the league had 771 players in total so little less than a third of the league's players were playing for a new team. This got us thinking if these frequent trades are productive at all, do they translate to success on the pitch, or do younger inexperienced teams with a high player turnover actually achieve less success.

In our research we want to answer a simple question. Do teams that let players stay on their squad for many consecutive years fare better, i.e. is there a correlation between success and how long the players of a football team stay together.

In an attempt to answer this hypothesis we decided to look at the results for the last 20 years in the most popular sporting league in the world, the English Premier League.

Project discussion

Research focus and scope

The English Premier League is our main and only focus for this research. Ideally we would have liked to look at multiple top leagues in the world of football. Taking it a step further would have meant the inclusion of other sports. Answering such questions as, “will basketball have a significantly different correlation?”, would definitely have been interesting. After doing some research we realised that we would have to write a good amount of data scrapers that would prove to be quite complex. We therefore narrowed our scope to only the Premier League but looked at a full 20 seasons spanning from 1997-’98 to the 2016-’17 season.

Clarification - In this report we refer to a team’s player stay. What this means is how long players have stayed with their team on average in seasons (1 year). The minimum togetherness age a player can have is 1 season, even though he might not have played a full season for the team.

Data gathering

Sources

Luckily the Premier League provides a decent website in which we could find all the information we desired to look at. We believe this to be an excellent source of data, it is in fact “straight from the horse’s mouth”. This included the standings for years 1998 to 2017 and more importantly the player data.

When starting out we thought 10 years of data would be sufficient, i.e. 2007 - 2017. We then realized that it was relatively easy for us to get data further back. That introduced one problem though. Premierleague.com only stored data from 1992, this is the year when the league was founded. Most if not all of the teams existed way before then[1]. What this meant for our data was that if we were to check how long a player had been with a team in 1992 it would return 1 year even though he had been there longer. Every player would have one

[1] https://en.wikipedia.org/wiki/List_of_football_clubs_in_England_and_Wales_by_year_formed

year on their record for their current team. This is why we felt like 1998 was a good starting year, even though there might be a few edge cases of players that got some years cut off. This cutoff would happen for every team equally so it should have little impact on the results when comparing the teams against each other.

Method

We wrote an automatic scraper for Premierleague.com in Python utilizing some useful Python libraries. To explain what data we extracted for each player we should take a look at what our scraper does:

For each season it looks at a specific team. It opens up the team's squad list and fetches the url for every player (https://www.premierleague.com/players). It then begins to open up the player profiles, one at a time. From the player profile it gathers how many seasons the player has been with said team and how many Premier League appearances he made on the pitch that season. This resulted in 16829 entries with said player information.

The next order of business was to extract something useful from this data. We needed to thin the herd a little bit. Why? Because the average team in the English Premier League has over 40 players on their squad list, many of whom only make a handful of appearances, are inactive or simply don't play any games for their club. To do this we wrote another cleanup script in Python which made a list of the 14 - 18 players that had the most appearances for their club at every given season (active squad), the cut-off for the amount being a player that made fewer than 15 appearances.

With this focused group of players that we refer to as the active squad, we calculated some useful data that could be viewed on a per team basis. Firstly the team's mean player stay (seasons at club) which gives us a kind of experience value for the team, a number that indicates the average number of seasons that the active squad has played together. A higher average player stays therefore possibly indicating a more unified squad.

We also calculated the median player stay, standard deviation of the mean player stay, the minimum and maximum seasons at the club and possible box plot values for where 50% of the players fit from the median value. After working with some visualizations we realized

we wanted an extra variable for each club, a value that told us if a team was entering the league during said season.

In addition to this data we needed the league results for the appropriate seasons. This was done manually, simply copying the 20 seasons into a separate document.

Visualizations

With the data in hand, we started by comparing the mean player stay for every club to their club's league position. Doing this we hoped that there would be a strong trend supporting our hypothesis.

Figure 1.

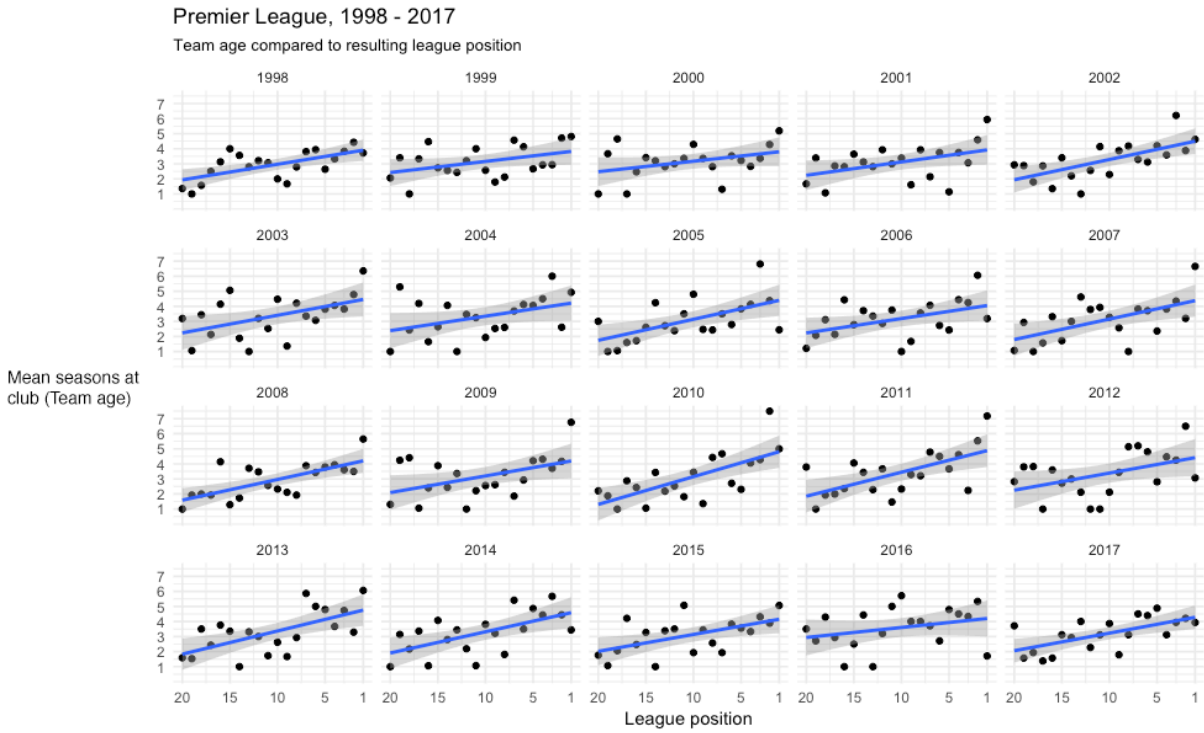


Figure 1 - Showing the correlation over 20 seasons in the premier league.

Calculating the correlation coefficient between the mean player stay at a club and their league position we found there to be a downhill sloping correlation of -0.53, downhill as the

higher the mean player stay the lower the league position number. A lower league position number actually translates as greater success as position 1 is the league winner but 20 is the relegated league loser. All our charts however reverse the axis for the league position so that the league winner is either at the top of the y-axis or right-most on the x-axis.

A correlation coefficient of 0.53 can be considered moderate, so there is definitely some tendency for more experienced teams (higher mean average stay) to perform better, but it's not strong and could be considered as an expected result in this context and not quite significant.

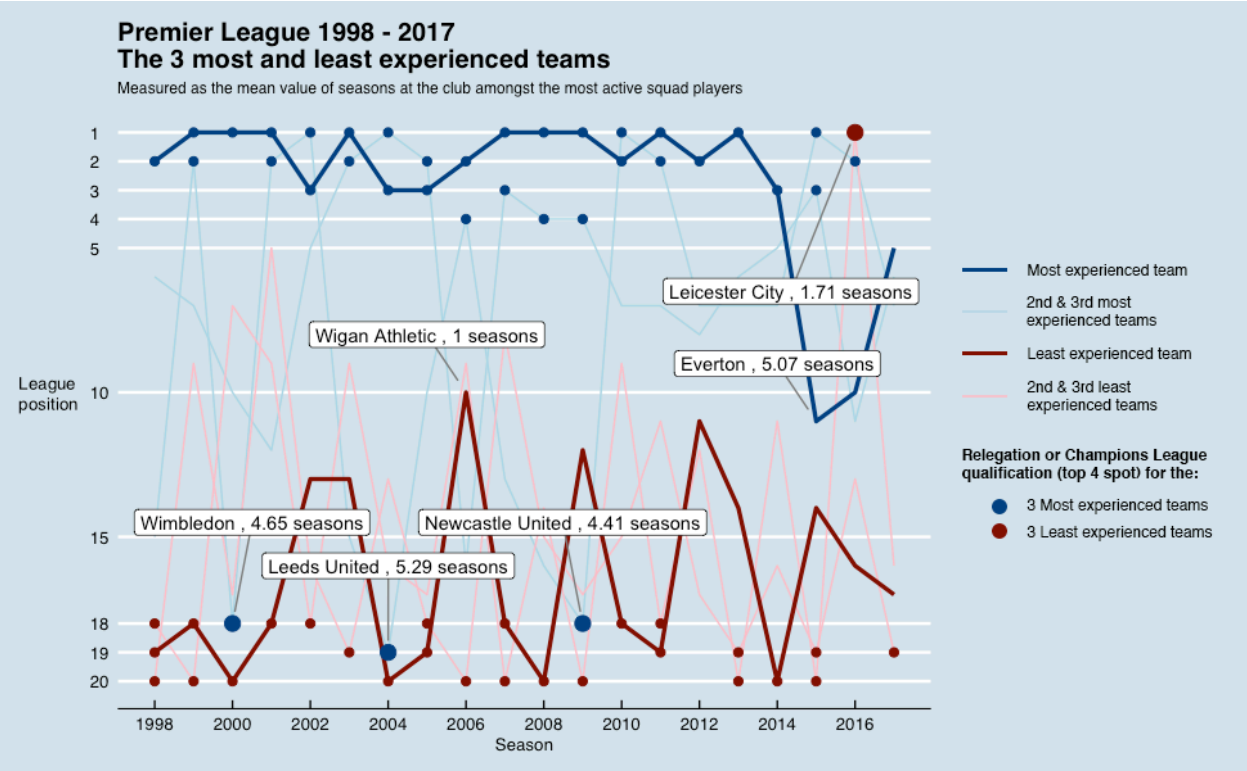
Looking at figure 1 with all our initial scatterplots, 1 for every season we see this sloping correlation where more experienced teams are sitting higher up the table. Furthermore we see that the oldest team is winning the league in 9 out of 20 seasons which we consider very significant and that the relatively youngest team is relegated 11 times in the 20 seasons.

We definitely wanted to visualize this better, as this was news to us and with further research we ended up comparing the 3 most experienced teams and the 3 least experienced teams for every season (fig2).

Only once in the last 20 years has 1 of the 3 least experienced teams been in the top 4 positions of the league, that was in 2016 when Leicester City surprised everyone, winning games weekend after weekend for a full season ending up winning the league, an extraordinary feat that probably won't be replicated by any team in the near feature.

It has happened 3 times that 1 of the 3 most experienced teams have been relegated, this happened famously to Leeds United in 2004 after 14 years in the top flight in the midst of financial troubles and to Wimbledon in 2000, also after 14 years in the Premier League. Newcastle was relegated in 2009 with a team age of 4.41 but they are the only one which has also returned to the Premier League since.

Figure 2.



When comparing the most and least experienced teams it is very interesting to see that the least experienced (youngest) team in any given season has never been higher up the table than 10th place. In the 2005-'06 season Wigan achieved this feat with a completely new team, where everyone of the most active squad members were playing their 1st season for the club according to our observations.

The 3 most experienced clubs are in any of the top 3 positions half of the time or 29 out of the 60 possible top 3 placements in 20 seasons as can be observed in figure 2. This is while the 3 least experienced teams are relegated (bottom 3) 26 times in 60.

Visualizing 20 years and thousands of players in a single chart

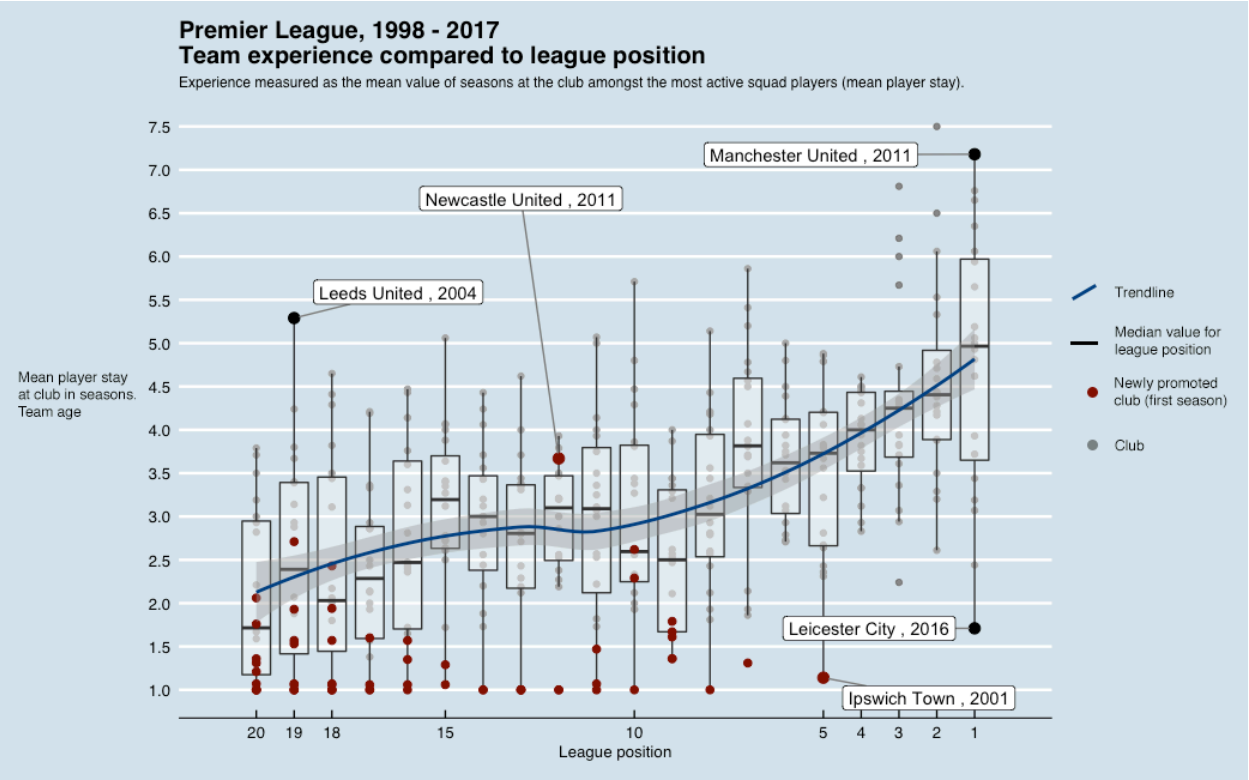
After visualizing the 3 relatively most and least experienced teams in figure 2 we tried to come up with a way to draft interesting information from all of the 20 seasons of data in a single chart. What we came up with can be seen in figure 3, a boxplot chart of the team age (mean player stay) in the 20 league positions over 20 seasons. Here you see the median

value and distribution of all the teams that have won the league or any other position in the league.

This is the plot that we are the most happy with and shares light on some very interesting information.

For the 80 teams that have finished in a top 4 position over the 20 seasons the median value is higher than of any other position in the league while the median values for the bottom 5 positions are lower than any other position in the league. This indicates a quite strong trendline where experienced teams tend to place at the top end of the league.

Figure 3.



The red dots in our chart indicate a team that was just promoted into the league. It is clear by looking at the chart that promoted teams do quite poorly. They also have a very low mean player stay. Our guess is that this is due to drastic squad changes with multiple player transfers happening when the club gets promoted to the Premier League. These red

dots are at the bottom of the chart as this is the first season for the average player in the team.

The 5 bubbles with club names and a year indicate an anomaly from the norm. Ipswich have the record over these seasons for finishing highest in the table during their first season after being promoted.

Newcastle United is the most experienced team during their first season after being promoted, this comes after being relegated in 2009 with one of the 3 most experienced teams as can be seen in figure 2 and not making too many changes in their lucrative battle to get back into the premier league.

Critique of our successes and failures

We believe that this research successfully showed a positive correlation between the mean player staying at a team and their success. This means that a team is more likely to succeed if their players stick together.

Even though we analysed the English Premier League as far back as their digital records allowed us we would have wanted to gather more information. We looked into gathering records for other top leagues such as the La Liga in Spain, the Bundesliga in Germany. In fact we had already manually scraped results from these leagues for 20 years back to match our findings in the Premier League. The problem was that we couldn't find information on each player in a form that allowed us to scrape it. The information could be found on different websites with different structures so that it simply would have been too much work.

Conclusion

With a correlation value of 0.53 there is a moderate positive correlation between teams that stay together for longer periods and success. Our visualizations show that there clearly is a relationship between players staying on teams longer and resulting in positive results. There might be an even stronger correlation in our last figure where the mean player stay is plotted in every league position over 20 seasons.